

V3 empirical study · Instruction

Two models · four underlyings · four volatility regimes · 1.5-year fixed calibration · option-implied Heston family

What this report is

- Companion to the seven-model 10,000-path fixed-calibration study. This report keeps only Heston and Heston–Merton, which are calibrated to listed calls (Fourier NLS). Rankings are within this pair; they are not compared here with return-based models.
- Companies are SPY, AAPL, MSFT, and AMZN. No new calibration or LSM. Every number is the stored cell from V3_1p5y_fixed_empirical_study.
- Return-based companion: V3_1p5y_fixed_empirical_study_return_based.pdf.
- Computational source of truth: the companion Jupyter notebook. This PDF is written from the same payload; every table number is identical.
- Question: which of 2 American-call pricing models tracks listed market prices most closely, and does the ranking change across companies and volatility regimes?
- In every table the BEST row is the lowest percentage RMSE. That row is bold and shaded green, and the Mark column says BEST.

Experimental design (held fixed for every model)

- Models: Heston (no jumps), Heston–Merton (Bates).
- Companies: SPY (primary), AAPL, MSFT, AMZN. Each name is calibrated and priced separately.
- Regimes (V3 one-year evaluation windows): Crisis 2008-08-01→2009-07-31; Normal 2014-01-01→2014-12-31; Late-cycle 2018-10-01→2019-09-30; COVID 2019-09-01→2020-08-31.
- The late-cycle and COVID files share September 2019 (V3 notebook dates). All other regime days are disjoint. Contracts are built independently in each window.
- Calibration window: 1.5-year ending at the first session of each 1-year evaluation window. Rolling: none (one estimate, held for the whole window). Parameters estimated at t_0 are used only after t_0 .
- If history is shorter than the requested lookback, the estimator uses all available observations up to the update date (equity starts 2003-12-01; listed-option panels start 2008-01).

V3 empirical study · Instruction (continued)

Two models · four underlyings · four volatility regimes · 1.5-year fixed calibration · option-implied Heston family

Shared market sample and filters (not model-specific)

- For each company $\times$ regime, one listed-call sample is drawn once and reused by all two models in this report: nearest-ATM call on each Monday (next session if Monday is closed), DTE 7–60, listed expiry, as-of that date.
- Estimation filters, applied in order to every quote used for Heston/Heston–Merton calibration and to the LSM comparison set: (1) calls only, finite S , K , C ; (2) no-arbitrage $C \geq \max(0, S-K)$; (3) $7 \leq \text{DTE} \leq 60$; (4) $|S/K - 1| \leq 10\%$; (5) premium ≥ 0.05 , valid bid-ask with relative spread $\leq 50\%$ when those columns exist, volume ≥ 1 when present.
- Return-based models (GBM, Merton, GARCH, GARCH–Merton) never see option quotes in §4; they still price the same filtered LSM contracts in evaluation.

Model-correctness adjustments (V3)

- Heston / Heston–Merton: $(\kappa, \theta, \xi, \rho, v_0)$ are option-implied Fourier NLS (Albrecher little-trap), not Method A realized-variance moments. μ is the P-measure lookback mean of stock returns; LSM replaces $\mu \rightarrow r$.
- Euler ordering: the stock increment over $[t, t+\Delta t]$ uses the current variance v_t ; $v_{t+\Delta t}$ is updated afterwards. No look-ahead in the variance.
- LSM: Longstaff–Schwartz American call on the full path cloud ($n_{\text{paths}} = 10000$, seed 42, $n_{\text{steps}} = \text{DTE}$, $\Delta t = 1/252$). Paths are not averaged before stopping.

Metrics

- Primary: $\text{RMSE\%} = 100 \times \sqrt{\text{mean}(((C_{\text{model}} - C_{\text{mkt}}) / C_{\text{mkt}})^2)}$. Ranking uses this number only.
- Secondary: $\text{MAE} = \text{mean} |C_{\text{model}} - C_{\text{mkt}}|$ (dollars). $\text{Bias \$} = \text{mean}(C_{\text{model}} - C_{\text{mkt}})$ (dollars). $\text{Bias\%} = 100 \times \text{mean}((C_{\text{model}} - C_{\text{mkt}})/C_{\text{mkt}})$. Positive bias = model expensive vs market. Early-exercise fraction = mean share of LSM paths that stop before expiry.

Page 3 · Filters, sample, and no-look-ahead

Shared evaluation sample · n listed calls (identical for all two models in this report)

Regime	Window	SPY n	AAPL n	MSFT n	AMZN n
2008-2009 Crisis	2008-08-01 → 2009-07-31	50	50	50	50
2013-2014 Normal	2014-01-01 → 2014-12-31	50	50	50	50
2018-2019 Late-cycle	2018-10-01 → 2019-09-30	50	51	50	51
2019-2020 COVID	2019-09-01 → 2020-08-31	50	50	50	50

Filter funnel · last-step n in the 1.5-year lookback ending at each regime (processed call panel)

Ticker	Regime	Input*	No-arb	DTE 7-60	S/K-1 ≤10%	Liquidity
SPY	2008-2009	3359	3311	3311	3203	3203
SPY	2013-2014	12815	12548	12548	12405	12405
SPY	2018-2019	35222	35104	35104	34809	34809
SPY	2019-2020	42100	41961	41961	41379	41379
AAPL	2008-2009	685	685	685	652	652
AAPL	2013-2014	8925	8875	8875	8633	8633
AAPL	2018-2019	6309	6297	6297	6084	6084
AAPL	2019-2020	8010	7992	7992	7713	7713
MSFT	2008-2009	566	566	566	538	538
MSFT	2013-2014	3299	3237	3237	3175	3175
MSFT	2018-2019	7814	7752	7752	7583	7583
MSFT	2019-2020	6892	6852	6852	6655	6655
AMZN	2008-2009	465	465	465	442	442
AMZN	2013-2014	5715	5692	5692	5598	5598
AMZN	2018-2019	28809	28808	28808	28346	28346
AMZN	2019-2020	26766	26760	26760	26323	26323

*Input is the processed call panel restricted to (regime end – 1.5-year, regime end]. Processed files already drop many raw quotes; the funnel above is the additional V3 research filter.

No look-ahead. Monthly parameters at update date t use only equity returns and listed quotes with $\text{trading_date} \leq t$. LSM prices the Monday contract using the latest calibration row with date $\leq$ that Monday. Quotes are as-of the grid timestamp (same session, else the prior session).

SPY 2008–2009 uses bid-ask mid when the source mark is \$0.01 (a GitHub field error, not missing trades). After that repair the processed panel has weekly coverage, so n matches AAPL/MSFT (~50). All two models in this report still share that exact sample.

Heston / Heston-Merton calibration further subsamples the filtered lookback surface to ≤ 24 contracts (maturity $\times$ moneyness strata). GBM / Merton / GARCH / GARCH-Merton estimate from lookback log returns only (jumps: 3σ residual threshold where applicable).

Page 4 · Table block · SPY · two models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=none, n_paths=10000, seed=42.

Table SPY-1 SPY · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	37.99	1.4513	-1.1865	-24.31	29.0%	2
Heston-Merton	37.76	1.4404	-1.1711	-23.83	27.2%	1

Table SPY-2 SPY · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	80.89	1.6010	+1.5789	+70.66	25.2%	1
Heston-Merton	80.98	1.6058	+1.5835	+70.80	25.2%	2

Table SPY-3 SPY · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	26.62	0.9285	+0.1483	+7.58	19.5%	2
Heston-Merton	23.17	0.9016	-0.3351	-3.49	11.7%	1

Table SPY-4 SPY · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	80.30	3.3486	+0.8238	+44.02	30.1%	1
Heston-Merton	86.96	3.5301	+1.2470	+51.47	23.5%	2

RMSE% is $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$. MAE and Bias \$ are in option-premium dollars. Bias% is $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$. Early-ex. frac. is the mean LSM share of paths e

Page 5 · Table block · Company summary · two models × four regimes

Each number is the arithmetic mean of AAPL, MSFT, AMZN. Same columns as the SPY page. Bold green row = lowest mean RMSE%. 1.5-year lookahead, rolling=none, n_paths=100

Table Co-1 Company summary · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · mean of AAPL, MSFT, AMZN · $\bar{n} = 50$

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	30.28	1.0825	-0.2769	-5.61	24.7%	2
Heston-Merton	30.05	1.0706	-0.3331	-6.76	19.1%	1

Table Co-2 Company summary · 2013-2014 Normal
2014-01-01 → 2014-12-31 · mean of AAPL, MSFT, AMZN · $\bar{n} = 50$

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	60.04	3.2210	+3.1840	+51.53	21.2%	2
Heston-Merton	55.52	2.7096	+2.5936	+46.54	19.0%	1

Table Co-3 Company summary · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · mean of AAPL, MSFT, AMZN · $\bar{n} = 51$

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	37.93	9.0136	+8.0834	+26.48	20.6%	2
Heston-Merton	33.51	7.8742	+6.5640	+20.76	20.3%	1

Table Co-4 Company summary · 2019-2020 COVID
2019-09-01 → 2020-08-31 · mean of AAPL, MSFT, AMZN · $\bar{n} = 50$

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	44.09	9.4461	-0.4458	+18.73	26.7%	1
Heston-Merton	49.63	10.7693	+5.5568	+25.44	20.3%	2

RMSE%, MAE, Bias \$, Bias%, and early-exercise fraction are each averaged across AAPL, MSFT, AMZN. This is not a pooled re-pricing of a combined contract sample. Ranking uses the mean RMS

Page 6 · Table block · AAPL · two models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=None, n_paths=10000, seed=42.

Table AAPL-1 AAPL · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	34.03	1.7307	-0.2640	+1.23	22.3%	1
Heston-Merton	34.10	1.7166	-0.2799	+1.17	19.5%	2

Table AAPL-2 AAPL · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	58.84	3.9883	+3.9883	+52.30	19.8%	1
Heston-Merton	58.99	3.9831	+3.9831	+52.19	17.1%	2

Table AAPL-3 AAPL · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 51 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	31.34	1.5900	+1.3922	+23.98	19.8%	2
Heston-Merton	21.47	1.0270	+0.5515	+10.59	22.7%	1

Table AAPL-4 AAPL · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	45.09	3.3073	+1.1657	+23.85	26.1%	1
Heston-Merton	45.11	3.3287	+1.1442	+23.69	20.6%	2

RMSE% is $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$. MAE and Bias \$ are in option-premium dollars. Bias% is $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$. Early-ex. frac. is the mean LSM share of paths

Page 7 · Table block · MSFT · two models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=none, n_paths=10000, seed=42.

Table MSFT-1 MSFT · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	27.04	0.3008	-0.2200	-14.34	27.4%	1
Heston-Merton	27.09	0.3016	-0.2193	-14.22	23.5%	2

Table MSFT-2 MSFT · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	62.26	0.4275	+0.4254	+52.75	24.5%	1
Heston-Merton	62.79	0.4308	+0.4271	+53.09	18.6%	2

Table MSFT-3 MSFT · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	28.71	0.7076	+0.4085	+15.07	22.6%	1
Heston-Merton	30.54	0.7557	+0.4887	+17.50	20.4%	2

Table MSFT-4 MSFT · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	47.96	1.7733	+0.1593	+19.55	25.7%	2
Heston-Merton	45.23	1.7365	-0.0483	+15.57	21.2%	1

RMSE% is $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$. MAE and Bias \$ are in option-premium dollars. Bias% is $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$. Early-ex. frac. is the mean LSM share of paths e

Page 8 · Table block · AMZN · two models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=none, n_paths=10000, seed=42.

Table AMZN-1 AMZN · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	29.77	1.2161	-0.3467	-3.72	24.3%	2
Heston-Merton	28.96	1.1935	-0.5002	-7.24	14.4%	1

Table AMZN-2 AMZN · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	59.01	5.2473	+5.1382	+49.55	19.3%	2
Heston-Merton	44.80	3.7149	+3.3705	+34.36	21.4%	1

Table AMZN-3 AMZN · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 51 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	53.74	24.7431	+22.4495	+40.39	19.5%	2
Heston-Merton	48.52	21.8398	+18.6520	+34.17	18.0%	1

Table AMZN-4 AMZN · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	39.20	23.2576	-2.6623	+12.80	28.2%	1
Heston-Merton	58.56	27.2428	+15.5745	+37.07	19.1%	2

RMSE% is $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$. MAE and Bias \$ are in option-premium dollars. Bias% is $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$. Early-ex. frac. is the mean LSM share of paths e

Page 7 · Summary tables · RMSE% across companies and regimes

Table S1 · Overall ranking (mean RMSE% over 16 cells)

Rank	Model	Mean RMSE%	Median RMSE%	# of 16 cells best
1	Heston-Merton	45.94	44.95	7
2	Heston	46.43	42.15	9

Table S2 · Wins by company (lowest RMSE% in that company×regime)

Model	SPY wins	AAPL wins	MSFT wins	AMZN wins	Total
Heston	2	3	3	1	9
Heston-Merton	2	1	1	3	7

Table S3 · Mean RMSE% by regime (equal-weight mean of SPY / AAPL / MSFT / AMZN)

Model	2008-2009 Crisis	2013-2014 Normal	2018-2019 Late-cycle	2019-2020 COVID	Mean of 4
Heston	32.21	65.25	35.10	53.14	46.43
Heston-Merton	31.98	61.89	30.93	58.97	45.94

Green row in S1 / S3 = lowest mean RMSE%. Green row in S2 = most company×regime wins. A model can win the most cells without having the lowest average error (and vice versa).

Table S4.1 · SPY

Model	Mean RMSE%	Median	# regimes best
Heston	56.45	59.15	2
Heston-Merton	57.22	59.37	2

Table S4.2 · AAPL

Model	Mean RMSE%	Median	# regimes best
Heston	42.33	39.56	3
Heston-Merton	39.92	39.61	1

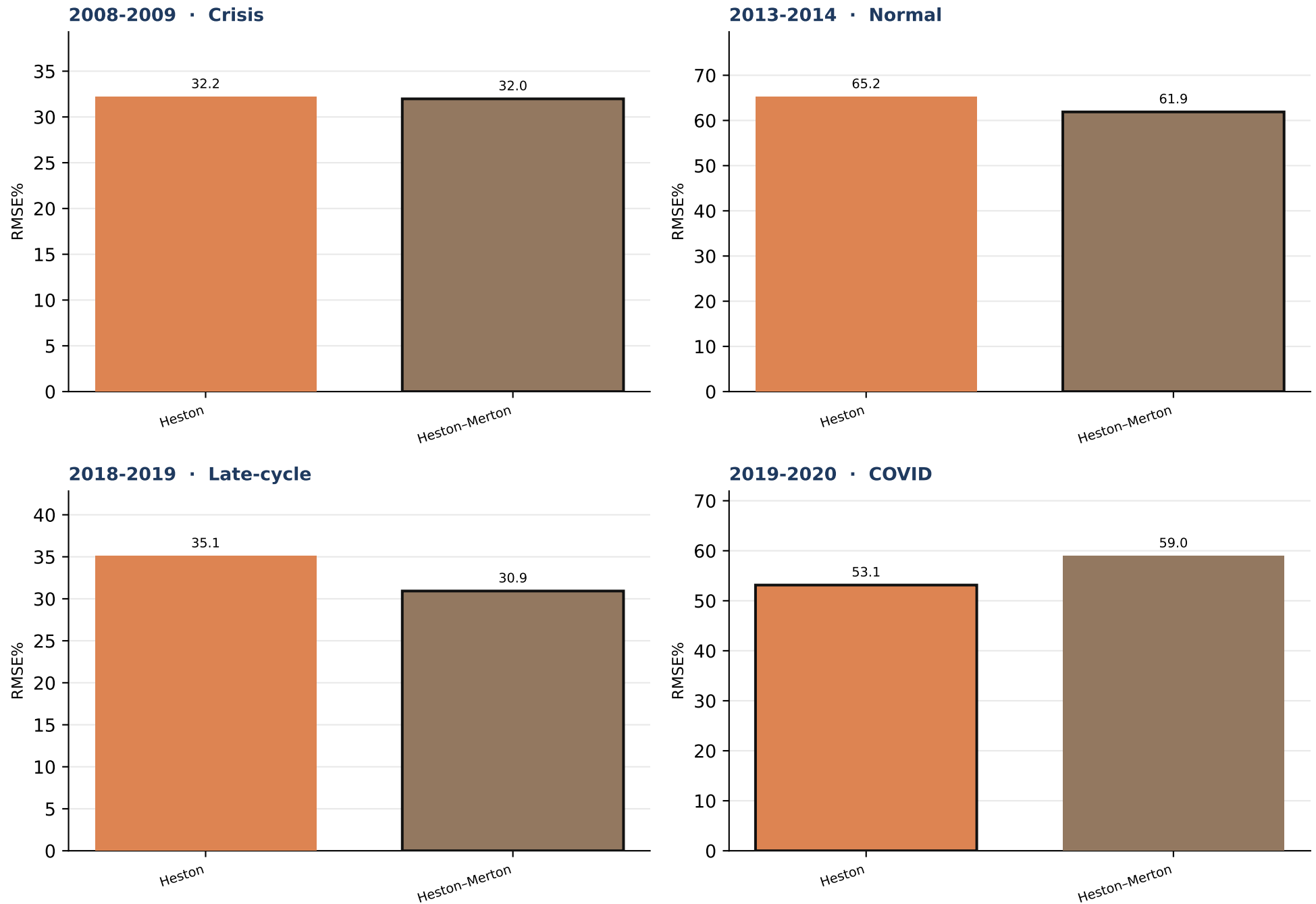
Table S4.3 · MSFT

Model	Mean RMSE%	Median	# regimes best
Heston	41.49	38.33	3
Heston-Merton	41.41	37.89	1

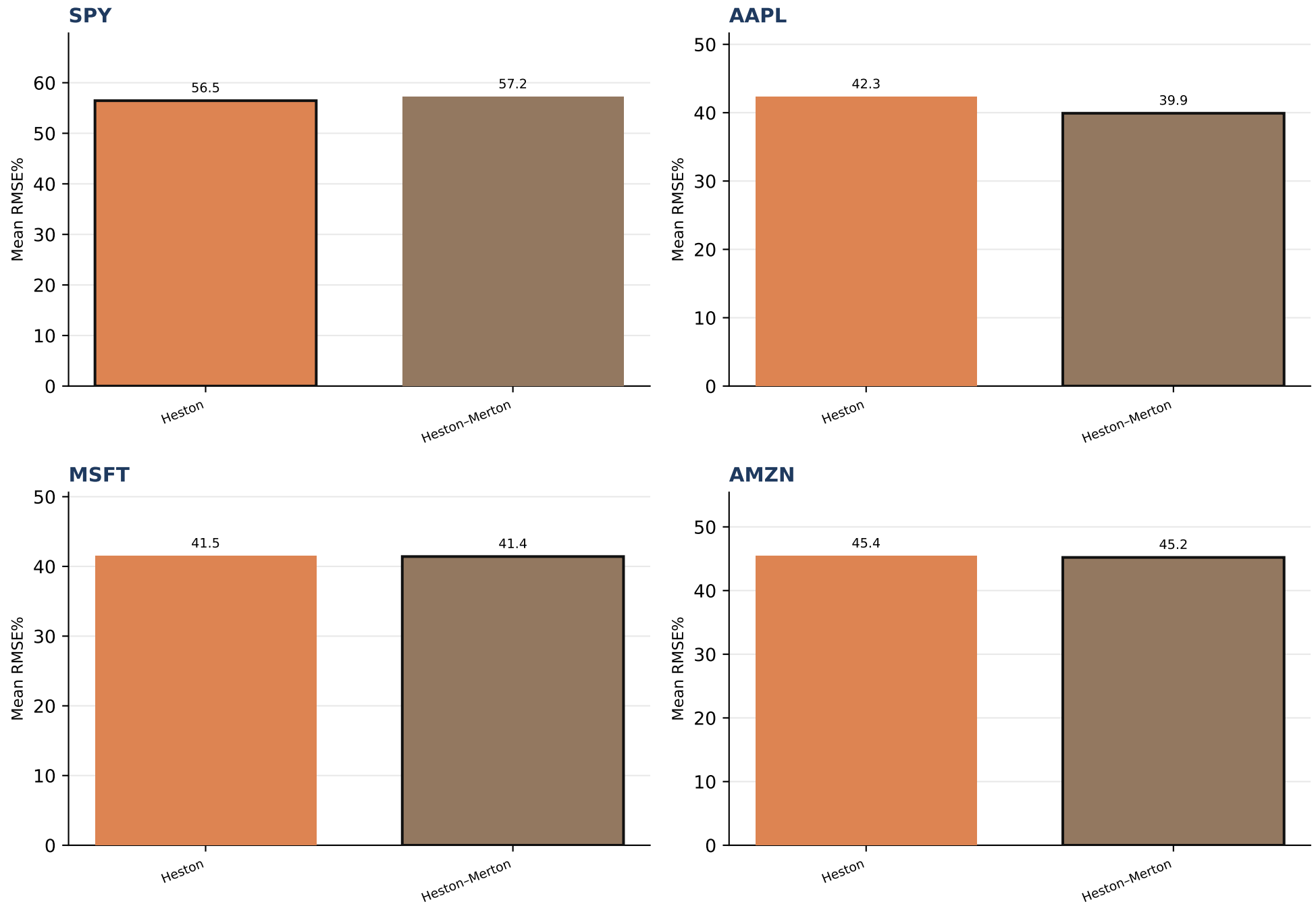
Table S4.4 · AMZN

Model	Mean RMSE%	Median	# regimes best
Heston	45.43	46.47	1
Heston-Merton	45.21	46.66	3

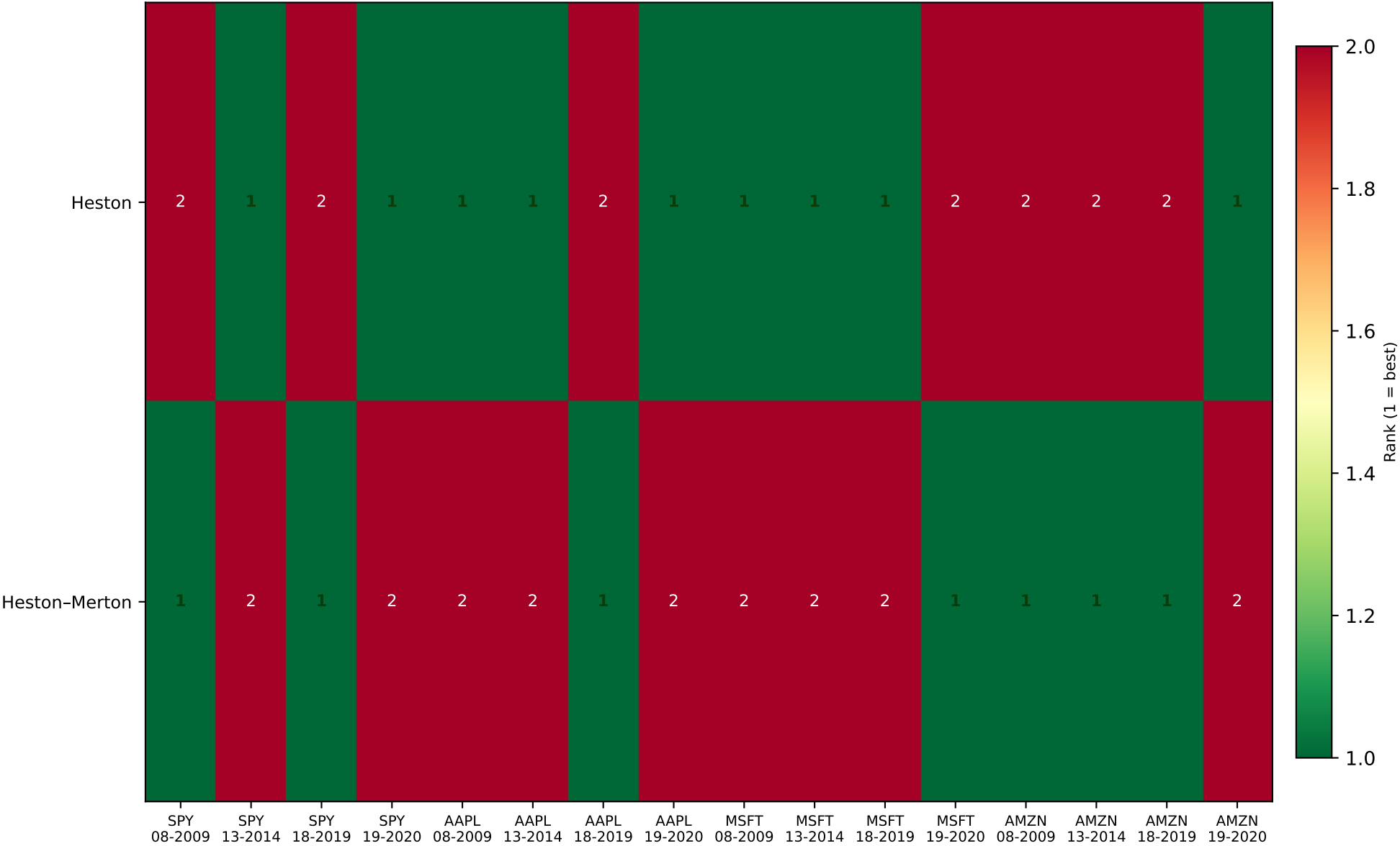
Each panel is one company. Green row = lowest mean RMSE% across the four V3 regimes for that name.



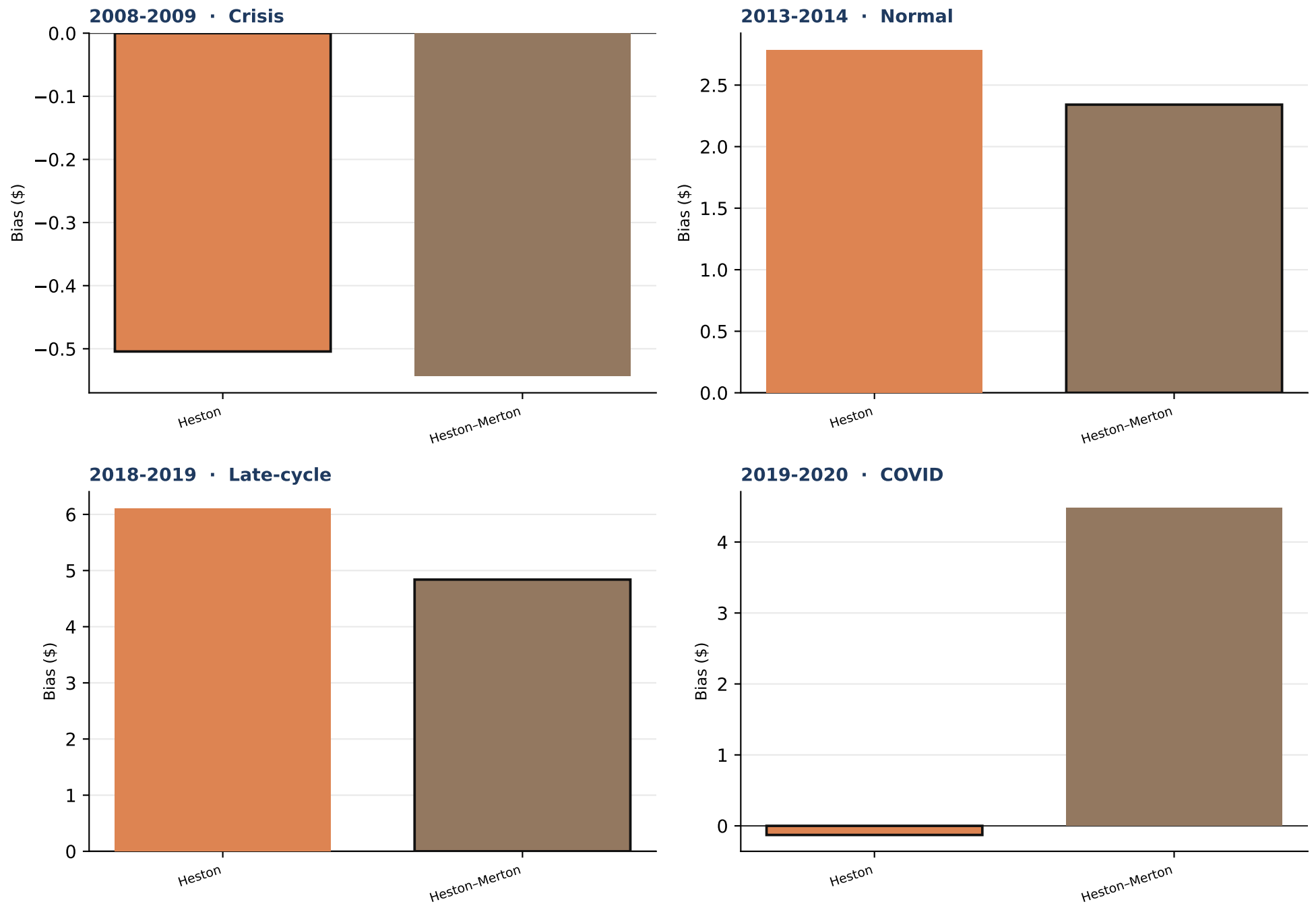
Outlined bar = lowest mean RMSE% in that regime. Equal-weight mean of the four companies.



Outlined bar = lowest mean RMSE% for that company.



Green / 1 = lowest RMSE% in that company×regime cell. Rankings can and do change across names and volatility states.



Outlined bar = closest to zero bias (not the RMSE ranking). Positive = model expensive vs listed quote.

Which models perform best, and does the ranking change?

- On mean percentage RMSE across the 16 company×regime cells, Heston-Merton is the best overall model (mean RMSE% = 45.94; 7 of 16 cells).
- Second overall is Heston (mean RMSE% = 46.43; 9 cells).
- The ranking is not stable. Best-in-cell models across the 16 tables: Heston, Heston-Merton.
- SPY: lowest mean RMSE% is Heston (56.45); wins 2 of 4 regimes.
- AAPL: lowest mean RMSE% is Heston-Merton (39.92); wins 1 of 4 regimes.
- MSFT: lowest mean RMSE% is Heston-Merton (41.41); wins 1 of 4 regimes.
- AMZN: lowest mean RMSE% is Heston-Merton (45.21); wins 3 of 4 regimes.
- 2008-2009 (Crisis): best mean RMSE% is Heston-Merton (31.98).
- 2013-2014 (Normal): best mean RMSE% is Heston-Merton (61.89).
- 2018-2019 (Late-cycle): best mean RMSE% is Heston-Merton (30.93).
- 2019-2020 (COVID): best mean RMSE% is Heston (53.14).
- Jump models as a group have lower mean RMSE% (45.94) than the no-jump models in this report (46.43).
- For the overall winner (Heston-Merton), crisis RMSE% is 31.98 vs 61.89 in the 2014 normal year — absolute pricing error is regime-dependent even when the ranking is not.
- These conclusions are conditional on the V3 filters, 1.5-year fixed calibration, Monday ATM sample, LSM with 10,000 paths, and American-call quotes only. They are not a statement about European options or other tenors.

How to read the 12 detailed tables

- Each of Tables SPY-1...MSFT-4 uses exactly the same option contracts for all two models in this report. Differences are the dynamics, not the sample.
- BEST is always lowest RMSE%. MAE, Bias \$, Bias%, and early-exercise are reported, not used for the ranking.
- A model that is BEST on RMSE% can still be biased (systematically expensive or cheap).
- Early-exercise fractions are a model implication, not an accuracy score; American calls on these